

PHAR6302 — Weeks 3 & 4 Study Notes

Dyslipidaemia and Type 2 Diabetes (Week 3), Weight Management (Week 4). Condensed from the course content, Therapeutic Guidelines, the Australian Medicines Handbook, the Heart Foundation lipid guide, the ADS Type 2 Diabetes Glycaemic Management Algorithm (May 2026), the Australian Obesity Management Algorithm and the Queensland Community Pharmacy clinical protocols. Verify anything assessment-critical against Canvas and the primary sources.

Week 3 · Dyslipidaemia

1. The governing principle

Lipid-lowering decisions are driven by **estimated absolute ASCVD risk**, not cholesterol numbers alone. The lipid profile tells you what to aim at; the risk category tells you whether and how hard to treat. Two exceptions override the calculator entirely: severe hyperlipidaemia (TC ≥ 7.5 or LDL-C ≥ 5.0 mmol/L) and abnormal lipids where risk can't be calculated (e.g. type 1 diabetes) — both go to the usual healthcare provider.

2. Targets by risk category (Therapeutic Guidelines)

Risk category	LDL-C	Non-HDL-C	Triglycerides
Established ASCVD / very high risk	$\geq 50\%$ reduction from baseline <i>and</i> < 1.4 mmol/L (whichever is lower)	< 2.2	< 1.7
High risk	$\geq 50\%$ reduction <i>and</i> < 1.8	< 2.6	< 1.7
Intermediate risk	< 2.6	< 3.4	< 2.0
Low risk	< 3.0	< 3.8	< 2.0

Very high risk includes: established ASCVD (CAD, ACS, stroke/TIA, PAD); multivessel CAD with $> 50\%$ stenosis or disease in more than one vascular bed; CKD with eGFR < 30 ; FH with atherosclerosis or a major risk factor; diabetes with end-organ damage or with ≥ 3 major risk factors; early-onset type 1 diabetes of more than 20 years' duration.

3. Reading a lipid profile

- **Ratios are out of favour.** TC:HDL and HDL:LDL get less emphasis now — focus on individual components (especially LDL-C) plus overall CV risk.
- **HDL is not simply "good cholesterol".** Interpret alongside triglycerides and clinical context, especially in type 2 diabetes. There is no evidence that raising HDL with drugs helps.
- **ApoB** — one molecule per atherogenic particle, so it counts particles rather than cholesterol mass; some regard it as a better risk marker than LDL-C alone.
- **Lp(a)** — LDL-like, largely genetic, barely shifts with diet or exercise; raises clot and blockage risk.
- **ApoA-1** — the main HDL protein; inversely related to ASCVD risk.

Fasting or not?

Non-fasting is acceptable for initial screening. Interpret cautiously if triglycerides exceed 4 mmol/L and collect a fasting sample if they're elevated. Fasting is preferred for tracking change over time. Always confirm an abnormal result with a repeat profile on a different day.

Screening and testing frequency

Screen from 45–79 (35 with diabetes; First Nations people 18–29 for individual risk factors, formal risk assessment from 30). Not on therapy: every 2 years at intermediate risk, every 5 years at low risk. **After starting or changing therapy: repeat the profile at 6 weeks** (AMH says 4–6 weeks during titration, then 6–12 months in maintenance).

4. Causes worth chasing before you prescribe

Secondary cause	Lipid effect
Hypothyroidism, nephrotic syndrome, cholestasis, anorexia nervosa	↑ LDL-C
Type 2 diabetes, obesity, renal impairment, smoking, drug therapy	↑ TG, ↓ HDL-C
High saturated fat diet	↑ LDL-C
Alcohol misuse, oestrogen use	↑ TG
Sedentary lifestyle	↑ LDL-C, ↓ HDL-C
Beta-blockers	↑ TC, ↓ HDL-C
Diuretics	↑ TC, ↓ TG

Primary (genetic) dyslipidaemia: familial hypercholesterolaemia — look for tendon xanthomata, corneal arcus before age 45, xanthelasma, and premature CHD in the family. Dutch Lipid Clinic Network criteria and cascade testing were covered in Week 1.

5. Lipid management pathway

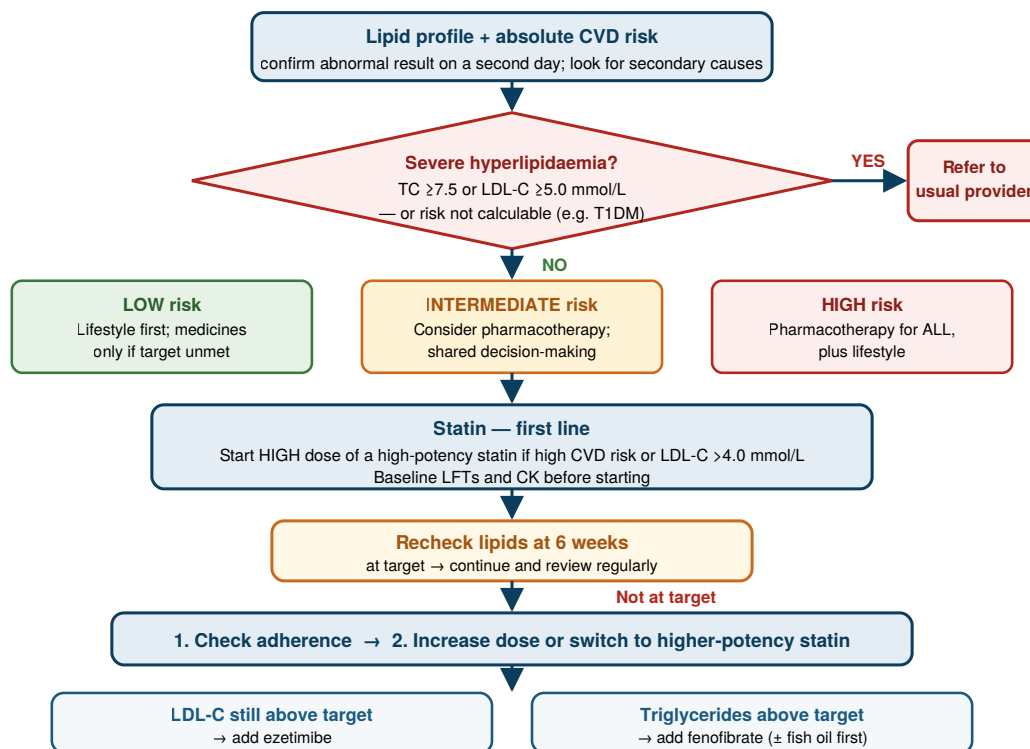


Figure 1. Lipid management pathway. PCSK9 inhibitors and bempedoic acid sit outside the current Queensland pharmacist-prescribing scope, but are covered in Tutorial 2 as emerging therapies — see section 10.

6. Statin intensity — the table to memorise

Intensity	LDL-C reduction	Daily dose
High	>50%	atorvastatin 40–80 mg; rosuvastatin 20–40 mg
Moderate	30–49%	atorvastatin 10–20 mg; pravastatin 40–80 mg; rosuvastatin 5–10 mg; simvastatin 20–80 mg*
Low	<30%	pravastatin 10–20 mg; simvastatin 5–10 mg

*Simvastatin 80 mg carries more myopathy risk than other statins achieving similar LDL lowering. Relative potency: atorvastatin and rosuvastatin > simvastatin > pravastatin. Dose increases at intervals of at least 4 weeks.

The 80% rule — why adding beats doubling

More than 80% of a statin's LDL-lowering effect is achieved at 50% of its maximum dose. Once you're there, adding ezetimibe (or low-dose colestyramine, or a sterol margarine) lowers LDL more than doubling the statin. Ezetimibe adds a further 15–25%. This is the pharmacological reason the escalation sequence in Figure 1 turns to combination therapy rather than pushing the statin dose indefinitely.

7. Statin safety — thresholds and rechallenge

Baseline: aminotransferase and CK. Repeat during treatment only if clinically indicated.

Stop the statin if:

- aminotransferase persistently **>3× ULN**
- **CK >10× ULN**
- persistent unexplained muscle pain or weakness **even with a normal CK**

If CK stays raised after stopping, look for another cause the statin may have unmasked (asymptomatic hypothyroidism, neuromuscular disease). **Rechallenge after at least 4 weeks** if the episode was mild and CK has normalised — consider a precipitant, a lower dose, or a different statin.

Counselling: report dark brown urine, or any muscle pain, tenderness or weakness.

Statin intolerance — don't accept the label too fast

Defined as inability to tolerate a statin at the dose needed for CV risk reduction, usually muscle symptoms. Fewer than 10% of patients genuinely experience adverse effects, yet **up to 75% of people started on a statin stop within 2 years** — and stopping increases major CV events.

Assess: timing of onset, response to withdrawal and rechallenge, symptom character, and alternative explanations (the placebo effect, other conditions, other medicines, unaccustomed activity are all common). Genuine intolerance is managed by dose reduction, switching statin, intermittent dosing, or non-statin therapy.

Never stop a statin during symptoms of ACS — cessation is associated with increased cardiac events, especially in the first week.

8. Worked target calculation

Baseline LDL-C 4.8 mmol/L. After 6 weeks on a high-potency statin, LDL-C is 2.9. Target is ≥50% reduction *and* <1.8 mmol/L.

Reduction achieved = $(4.8 - 2.9) \div 4.8 = \sim 40\%$. A 50% reduction would mean reaching 2.4 or below, and the absolute target is 1.8. **Target not met on either limb.** Check adherence first, then increase the dose or switch, then consider adding ezetimibe.

9. Drug-class monitoring

Drug	Monitoring
Statins	LFTs and CK at baseline; retest if clinically indicated. Lipids at 6 weeks after start or change, then 6–12 monthly.
Fenofibrate	FBC and LFTs before starting and 3-monthly through the first year; CK before starting and regularly if combined with a statin; monitor renal function and reduce dose if CrCl 30–60 mL/min. Watch for a paradoxical HDL fall, which can appear weeks to years in — consider stopping if HDL drops markedly from baseline.

Week 3 lipid nuggets — statins and standard therapy

- **Gemfibrozil is the fibrate to avoid with a statin** — fenofibrate is the one used in combination, which is why the Queensland protocol names it specifically.
- Statins are CYP (mainly 3A4) and OATP1B1 substrates. **Pravastatin and rosuvastatin have fewer CYP interactions** — useful when an interacting drug can't be avoided. Simvastatin caps: 10 mg with amlodipine, diltiazem or verapamil; 20 mg with amiodarone.
- **Sodium fusidate**: stop the statin for the duration and about a week afterwards — rhabdomyolysis fatalities have been reported.
- Fibrates are for **severe hypertriglyceridaemia (TG >10 mmol/L) to prevent pancreatitis**. Statins are not used for that purpose.
- Diet specifics for dyslipidaemia: swap saturated and trans fat for mono/polyunsaturated (avocado, nuts, seeds, vegetable oils); soluble fibre (oats, psyllium, legumes); plant sterol products; soy protein; **2–3 servings of oily fish a week**. Refer intermediate and high-risk patients to a dietitian.

10. Queensland Lipid Modification Protocol V2 — the scope boundaries

Element	Criteria
Eligible	Ages 18–79 with abnormal lipids — new or existing dyslipidaemia
Ineligible	Under 18 or over 79; pregnant or planning; CKD stage 3–5 ; familial hypercholesterolaemia ; type 1 diabetes; microvascular complications; complex CVD (severe hypertension $\geq 180/110$, congenital or rheumatic heart disease, heart failure, arrhythmias, PAD); severe respiratory disease; prior cardiothoracic surgery, ACS, stroke or hypertensive emergency; current DVT/PE or anticoagulant therapy; specialist care within 12 months without referral; severe lipid elevation (TC ≥ 7.5, LDL ≥ 5.0, TG ≥ 6) ; severe hyperglycaemia (HbA1c $\geq 10\%$, BGL ≥ 20) or hypoglycaemia (<4); not at lipid target after 6 months of therapy
May prescribe	Initiate or adjust dual combinations of statin (first-line; high-dose high-potency if high CVD risk or LDL >4.0), ezetimibe (monotherapy if statin contraindicated, or as add-on) and fenofibrate (add-on for triglyceride target). May modify existing regimens, ideally with the original prescriber
Immediate referral	BP $\geq 180/110$; HbA1c $\geq 10\%$ or BGL ≥ 20 ; TC ≥ 7.5 , LDL ≥ 5.0 or TG ≥ 6 ; eGFR <60 or uACR ≥ 3 ; any new ineligibility criterion; not at lipid target after 6 months
Monitoring	Baseline CVD risk category, lipid profile, LFTs and CK before starting; 6-week review of response to new or changed therapy; fenofibrate needs 3-monthly monitoring initially. CVD Risk Management Plan updated and sent to the usual provider each visit

The FH trap. Familial hypercholesterolaemia is on the *ineligible* list, not the treat-and-concurrently-refer list. So a patient with tendon xanthomata, corneal arcus before 45, or TC ≥ 7.5 / LDL ≥ 5.0 suggesting FH is **referred, not managed** — even though you'd manage a patient with prior ACS or on anticoagulants under a concurrent-referral arrangement in the hypertension protocol. Know which list each condition sits on.

Three protocols, three different "not at target" intervals. Hypertension: 4–6 weeks on two maximum-dose agents. Blood glucose: 3 months on two maximum-dose agents. Lipids: **6 months**. These are easy to conflate and are exactly the sort of detail a written paper targets.

11. Emerging lipid-lowering therapies (Tutorial 2)

These sit **outside the Queensland pharmacist-prescribing protocol**, but Tutorial 2 covers them explicitly — the learning outcome is to describe emerging lipid-lowering treatments and their potential impact on future practice. Expect to know mechanism, magnitude of effect and place in therapy rather than prescribing detail. Patients also ask about them, which is reason enough to be fluent.

Agent	Mechanism	LDL effect	Administration	Place in therapy
Evolocumab	PCSK9 inhibitor — monoclonal antibody preventing LDL-receptor degradation, so more receptors clear LDL	↓ 60% alone or with a statin; ↓20–30% in homozygous FH; also raises HDL	SC every 2 or 4 weeks	Reduces ischaemic CV events when added to a statin in established CVD. An option in familial hypercholesterolaemia, usually with a maximally tolerated statin. Well tolerated; injection site reactions
Inclisiran	Small interfering RNA — inhibits hepatic <i>production</i> of PCSK9 rather than blocking the circulating protein	↓ 50% with or without a statin	SC every 6 months after the first two doses; given by a health professional	An option when evolocumab is inappropriate. CV morbidity and mortality outcomes not yet determined — the LDL lowering is established, the outcome benefit is not
Bempedoic acid	ATP-citrate lyase inhibitor — acts upstream of HMG-CoA reductase in the same pathway, but is activated only in the liver, not muscle	Modest; used as add-on	Oral	Not in the Queensland guideline scope. The liver-selective activation is the rationale for use in statin intolerance

Why the dosing intervals differ — a good exam discriminator

Evolocumab is an antibody that binds circulating PCSK9, so its effect lasts as long as the drug does: fortnightly or monthly. Inclisiran silences the messenger RNA that makes PCSK9, so hepatic production stays suppressed long after the injection — hence **twice a year**. Same target, different level of intervention, very different adherence profile. That six-monthly schedule administered by a health professional is a plausible future pharmacist role even though prescribing is not.

Lp(a) connects here. Lp(a) is genetically determined and barely responds to lifestyle or statins; PCSK9 inhibitors lower it modestly, and dedicated Lp(a)-lowering agents are in development. This is why Tutorial 1's Lp(a) content and Tutorial 2's novel therapies content belong together.

Week 3 · Type 2 Diabetes

1. Diagnosis (Therapeutic Guidelines Table 18.2)

Test	Diabetes	Prediabetes (adults)
HbA1c	≥48 mmol/mol (6.5%)	42–46 mmol/mol (6.0–6.4%)
Fasting venous glucose	≥7.0 mmol/L	6.1–6.9 (impaired fasting glucose)
Random venous glucose	≥11.1 mmol/L	none defined; investigate further if 7.8–11.0
OGTT, 2 h after 75 g	≥11.1 mmol/L	7.8–11.0 (impaired glucose tolerance)

- **Symptomatic patient:** one positive test confirms. **Asymptomatic:** repeat on a different day, preferably the same test.
- Venous laboratory samples only — **do not diagnose on finger-prick testing.**
- If two tests disagree, repeat the one above threshold and diagnose on that.
- An HbA1c below 48 mmol/mol does not refute a diagnosis made on venous glucose or OGTT.
- Established diagnosis needs a fasting glucose within the past month and an HbA1c within the past 3 months.

HbA1c can mislead — know when not to trust it

Falsely low: any haemolytic anaemia, severe kidney impairment, severe liver disease, anaemia of chronic disease, regular phlebotomy (haemochromatosis, blood donors), recent transfusion, pregnancy and **the first 3 months postpartum** (don't use it at all in that window). **Falsely high:** iron deficiency. Haemoglobinopathies can interfere with the assay.

If HbA1c and blood glucose disagree, suspect an interfering condition and use venous glucose instead.

2. Glycaemic targets

Target	Who
≤53 mmol/mol (7.0%)	General target for most adults, including longer duration or established CVD, without high hypoglycaemia risk
Tighter (e.g. ≤6.5%)	Newly diagnosed, short duration, lifestyle ± metformin only, low hypoglycaemia risk, no significant CVD; also young-onset T2DM aged 18–30
Less stringent (~64 mmol/mol, 8%)	Prior severe hypoglycaemia, limited life expectancy, advanced complications, extensive comorbidity, long-standing diabetes not reaching target despite intensive therapy

Other targets: fasting glucose 4–7 mmol/L; postprandial 5–10 mmol/L (Diabetes Australia gives 6–8 fasting and 6–10 postprandial for self-monitoring).

HbA1c reflects the preceding 2–3 months. Monitor **every 3 months if not at target or adjusting therapy; every 6 months if stable and at target.**

Unit conversion trap. 48 mmol/mol = 6.5% (the diagnostic threshold). 53 mmol/mol = 7.0% (the treatment target). Don't pair 6% with 48 mmol/mol — they aren't equivalent.

Targets are a floor to individualise from, not fixed categories. A patient already below their band on well-tolerated therapy with no hypoglycaemia may reasonably aim lower still — the guideline principle is that *further improvement is appropriate if it can be achieved safely*. Equally, intensive control is inappropriate where hypoglycaemia risk is unacceptable. Don't read "<7.0% on dual therapy" as a rule that stops further optimisation; ask what is safely achievable for this person.

3. Assessment beyond the HbA1c

Signs and symptoms

Insulin resistance / PCOS: acanthosis nigricans, skin tags, central obesity, hirsutism. **Poor control or complications:** lethargy, polyuria, polydipsia, recurrent fungal or bacterial infection, blurred vision, poor wound healing. **Red flags:** unexplained weight loss or gain, frequent headache, dizziness, chronic nausea or vomiting.

Pathology to review

HbA1c, fasting glucose, UEC and eGFR, **urine ACR**, lipid profile, LFTs (MASLD is common), FBC (anaemia distorts HbA1c), **vitamin B12 if on metformin**, urinalysis, last eye and foot check.

MASLD — worth knowing

Metabolic dysfunction-associated steatotic liver disease (formerly NAFLD), with MASH the severe form, is a leading cause of fibrosis and cirrhosis and is often silent. **ALT rises first and may only be modest — above 30 U/L can already indicate injury** in someone with metabolic risk, since traditional reference ranges are considered too high. Waist-to-height ratio ≥ 0.5 correlates strongly with the visceral fat driving it. Ultrasound is first-line for steatosis, but **FIB-4 and elastography are needed to stage fibrosis** — ultrasound alone can't.

Cardio-renal-metabolic framing: CVD, CKD and metabolic disease share pathways, and SGLT2 inhibitors, GLP-1 agonists, tirzepatide and finerenone all act across that cycle.

4. The stepped treatment algorithm

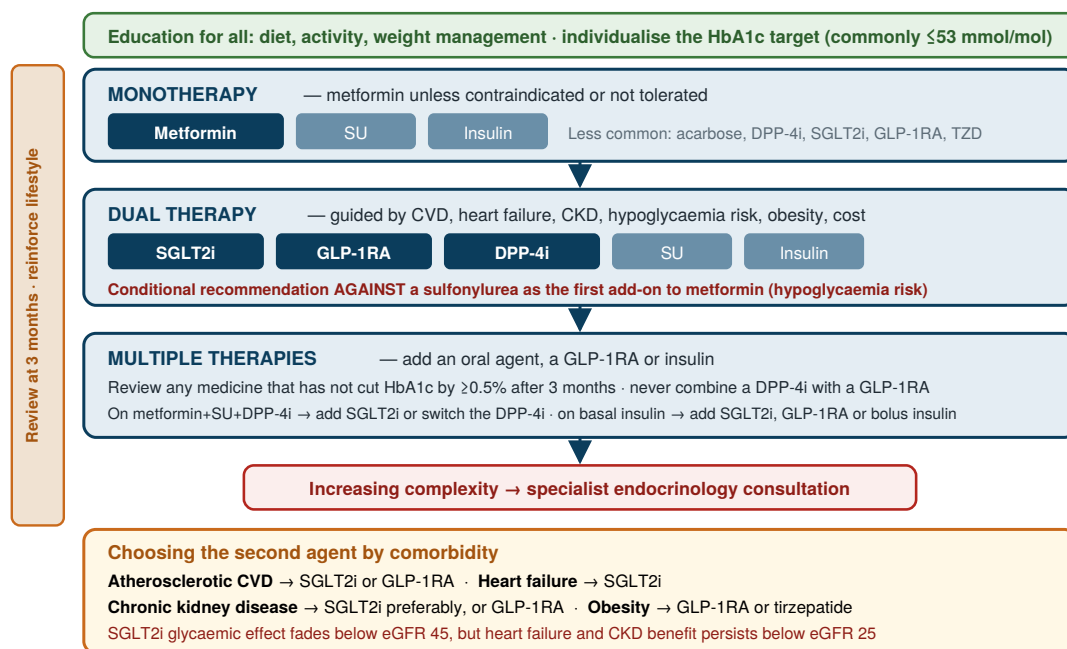


Figure 2. Australian Type 2 Diabetes Glycaemic Management Algorithm (ADS, May 2026). GLP-1RA with SGLT2i is not a PBS-approved combination for type 2 diabetes.

5. The drug classes at a glance

Class	HbA1c fall added to metformin	Weight	Hypoglycaemia	Key points
Metformin	0.8–1.1% (monotherapy)	neutral or ↓	no	First line. 500 mg 1–3 times daily up to 850 mg 2–3 times daily, max 3 g. Contraindicated eGFR <30 ; dose ceilings 2 g / 1 g / 500 mg at CrCl 60–90 / 30–60 / 15–30. GI effects, B12 malabsorption, rare lactic acidosis
SGLT2 inhibitors	0.4–0.7%	↓	low alone	CV and renal benefit. Genital candidiasis, balanitis, UTI, euglycaemic ketoacidosis . Stop 3 days before surgery. Early creatinine rise of 3–5 mL/min is expected
GLP-1 analogues	0.7–1.5%	↓	rare alone	CV benefit. Nausea/vomiting in up to 50%, usually settles. Delayed gastric emptying — aspiration risk at surgery or endoscopy . Avoid with pancreatitis history
DPP-4 inhibitors	0.5–0.6%	neutral	no	No CV benefit; saxagliptin increased heart failure hospitalisation. Linagliptin needs no renal dose adjustment. Never combine with a GLP-1RA
Sulfonylureas	0.5–0.7%	↑	common	Glibenclamide highest hypoglycaemia risk; gliclazide and glipizide lower and preferred in renal impairment. Take with food
Tirzepatide (GLP-1/GIP)	2–2.5%	↓	rare alone	Largest HbA1c fall of the oral/injectable agents. GI effects very common but usually transient
Insulin	0.7–1.2% or more	↑	common	Reduction theoretically unlimited with titration. Usually started as once-daily basal (e.g. glargine)

GLP-1 regimens for type 2 diabetes

Drug	Renal floor	Regimen
Dulaglutide	GFR ≥15	1.5 mg SC once weekly — no titration
Liraglutide	GFR ≥15	0.6 mg daily for 1 week → 1.2 mg; after ≥1 week, up to max 1.8 mg daily
Semaglutide	GFR ≥30	0.25 mg weekly for 4 weeks → 0.5 mg; after ≥4 weeks, up to max 1 mg weekly

Once-weekly agents take **6–8 weeks to reach maximum effect** — don't judge response too early. Never withdraw a GLP-1 from the pen into a syringe. Note these diabetes doses are well below the weight-management doses (semaglutide 2.4 mg weekly, liraglutide 3 mg daily).

6. Sick-day management

WITHHOLD Metformin dehydration → ↓clearance → lactic acidosis; worsens nausea SGLT2 inhibitors illness + dehydration + low carbohydrate → euglycaemic DKA Restart both only when eating and drinking normally and renal function is	CONTINUE (with conditions) DPP-4 inhibitors continue — unlikely to cause hypoglycaemia Sulfonylureas continue as tolerated; withhold if glucose trending low GLP-1RA — withhold only if it is worsening nausea/vomiting
Go straight to hospital if any of these appear - severe or worsening hyperglycaemia - persistent vomiting more than 4 hours - high fever - abdominal pain - severe headache - drowsiness - marked ketosis (blood ketones >1.5 mmol/L = impending DKA) - recurrent hypoglycaemia Blood ketones ≥0.6 mmol/L with hyperglycaemia needs urgent advice - a normal glucose does NOT exclude DKA on an SGLT2 inhibitor	

Figure 3. Sick-day adjustment of non-insulin agents (Therapeutic Guidelines). All type 1 and most type 2 patients should have an individualised written sick-day plan.

Numeric triggers: not on insulin — seek help if the last two glucose readings were above 15 mmol/L. On insulin — seek help early; if glucose is above 12 without vomiting or hypos, give the usual dose plus 10–20% (110–120% of usual) while glucose stays high. Higher DKA risk in type 2 diabetes if severely unwell, underweight, markedly carbohydrate-restricted, pregnant, previously ketotic, or on an SGLT2 inhibitor.

7. Queensland Blood Glucose Management Protocol V2 — the scope boundaries

Element	Criteria
Eligible	Ages 18–79 with new or existing type 2 diabetes and no exclusions
Ineligible	Pregnant or planning; CKD stage 3–5; type 1 diabetes; insulin-managed type 2 diabetes ; diabetic complications (retinopathy, neuropathy, nephropathy); severe hypertension $\geq 180/110$; complex CVD (heart failure, arrhythmias, AF, congenital or rheumatic heart disease); recent cardiothoracic surgery, ACS or stroke; anticoagulant use or thromboembolic disease; recent specialist care without referral back; severe hyperglycaemia (HbA1c $\geq 10\%$ or BGL ≥ 20) or hypoglycaemia; extreme lipid derangement; uncontrolled asthma or COPD
May prescribe	Initiate or adjust up to two agents from metformin (first-line), SGLT2 inhibitors, GLP-1 receptor agonists and DPP-4 inhibitors — chosen on renal function, CVD risk, age, comorbidities and preference
Referral triggers	Inadequate response after 3 months on two maximum-dose agents ; severe hyper- or hypoglycaemia; development of nephropathy, neuropathy or retinopathy; uncontrolled comorbidities; intolerable adverse effects
Monitoring	First review 3 months after initiation for HbA1c response; then every 3 months if unstable or 6 months if stable and at target; annual cycle of care. CVD Risk Management Plan to the usual provider each visit

Note the contrast with the lipid protocol: pharmacists can prescribe **two** glucose-lowering agents but the patient must not already be on insulin, and any microvascular complication is an outright exclusion rather than a reason for shared care.

8. Monitoring and the Annual Cycle of Care

Drug-specific

- **Metformin:** eGFR before starting and every 4–6 months; **vitamin B12 every 12 months**
- **SGLT2i:** eGFR and creatinine before starting and periodically
- **GLP-1RA:** renal function before starting and periodically, especially if at risk of volume depletion
- **DPP-4i:** eGFR and creatinine before starting and periodically; most need renal dose adjustment (not linagliptin)

Annual Cycle of Care

- **HbA1c:** 3-monthly if unstable, 6-monthly if stable
- **Blood pressure:** every review
- **Feet:** every 1–12 months by risk category
- **Eyes:** every 2 years, more often if retinopathy
- **Dental:** every 12 months
- **uACR, eGFR, lipids:** every 12 months
- **Weight and waist:** every 6 months
- **Nutrition, activity, medication review, smoking, psychosocial:** every 12 months

Week 3 diabetes nuggets

- **The 0.5% rule:** if an added agent hasn't cut HbA1c by at least 0.5% after 3 months, consider stopping it and trying something else. If targets aren't met on two agents at adequate doses, escalate or refer.
- **Weight loss $\geq 10\%$ may allow reduction or cessation of glucose-lowering medication;** even $>5\%$ improves control and can produce remission, especially within 6 years of diagnosis.
- Combining weight-loss interventions with diabetes drugs needs care: **SGLT2 inhibitors plus a very-low-carbohydrate or ketogenic diet risks euglycaemic ketoacidosis**, and insulin or sulfonylureas plus reduced intake risks hypoglycaemia.
- **Drugs that raise glucose:** glucocorticoids, antipsychotics, calcineurin inhibitors, somatropin, high-dose thiazides. Beta-blockers mask hypoglycaemia warning signs; alcohol lowers glucose and masks warnings too.
- Physical activity for diabetes: 150 min/week moderate-vigorous aerobic **plus two resistance sessions**.
- Semaglutide has a **retinopathy signal** in patients with existing retinopathy also using insulin — unclear whether it's the drug or the speed of glycaemic improvement.

Week 4 · Weight Management

1. Framing obesity correctly

WHO defines obesity as abnormal or excessive fat accumulation presenting a health risk. About **67% of Australian adults** live with overweight or obesity, and **over 70% of First Nations adults** — where excess weight accounts for nearly 10% of total disease burden and is a leading contributor to the health gap.

Obesity is a complex chronic disease, not a lifestyle choice: dysregulated hypothalamic appetite and satiety pathways; impaired leptin, insulin, ghrelin and peptide YY signalling; leptin resistance despite high leptin; adipose tissue acting as an endocrine organ releasing adipokines and inflammatory cytokines that drive insulin resistance; altered gut microbiota, brain reward pathways and adaptive falls in energy expenditure. That biology is why weight is regained when treatment stops, and why framing obesity as personal failure is both wrong and counterproductive.

2. Talking about weight without doing harm

Language

Use: person with obesity, person living with obesity, person with overweight, higher or excess body weight, healthier weight range.

Avoid: obese person, overweight person, morbidly obese, weight problem, fat, heavy — and anything implying blame, laziness or lack of willpower.

Ask permission first: "Would it be okay if we spent a few minutes talking about how your weight may be affecting your health?" Respect a decline and leave the door open.

Weight stigma from past healthcare encounters drives lower self-esteem, anxiety, depression, disordered eating, healthcare avoidance and worse outcomes.

Match the conversation to readiness

- **Pre-contemplation:** validate experience, raise awareness gently, no pressure, leave the door open
- **Contemplation:** explore pros and cons, strengthen motivation, encourage reflection
- **Preparation:** reinforce motivation, identify achievable first steps, problem-solve barriers, build support

Do: listen more than you speak, ask open questions, explore barriers without judgement, acknowledge social and economic influences, set specific behavioural targets. **Don't:** lecture, assume low motivation, focus only on BMI, imply weight loss is easy, or start a sentence with "you just need to..."

Celebrate non-scale wins: blood pressure, glucose, mobility, energy, sleep, quality of life.

3. Measuring and classifying

BMI (kg/m ²)	Classification
18.5–24.9	Healthy weight
25.0–29.9	Overweight
30.0–34.9	Obesity class I
35.0–39.9	Obesity class II
≥40.0	Obesity class III (severe)

BMI doesn't distinguish fat from lean mass or show fat distribution — always read it alongside central adiposity and clinical risk. Population-specific cut-offs are lower for Asian and Indigenous Australian populations (27.5–37.5 versus >37.5 in the Australian Obesity Management Algorithm).

Waist circumference — the better predictor

Measured at the **midpoint between the lower rib margin and the iliac crest, after normal expiration, standing.**

	Increased	Substantially increased
Men	≥94 cm	≥102 cm
Women	≥80 cm	≥88 cm

Lower thresholds apply for some Asian populations. **Waist-to-height ratio ≥0.5** indicates increased risk and needs no ethnic adjustment — the rule of thumb is that your waist should be less than half your height. DXA isn't routine but is increasingly useful for tracking lean mass loss during pharmacotherapy.

4. Screening for complications

Domain	Tool and referral trigger
Obstructive sleep apnoea	STOP-BANG — refer if a risk factor plus multiple symptoms and/or score >3
Mental health	Kessler K10 — refer if score >20
Eating disorders	SCOFF or InsideOut Screener — refer if suspected
Diabetes	AUSDRISK every 3 years if over 40; look for insulin-resistance signs. HbA1c/FBG if no test in 3 years plus a risk factor. Emergency referral if HbA1c >10% (86 mmol/mol) or any BGL >20 mmol/L with ketosis, dehydration, vomiting, infection or altered consciousness
Dyslipidaemia	AusCVD risk tool plus lipid panel; refer for FH signs (tendon xanthomata, arcus before 45, xanthelasma)
Thyroid	Test if hypothyroid symptoms; refer if TSH >4 mIU/L
Cardiometabolic and musculoskeletal	Hypertension, PVD, CVD; osteoarthritis, reduced mobility, chronic pain

Health risks of obesity include type 2 diabetes, CVD, stroke, hypertension, dyslipidaemia, OSA, osteoarthritis, fatty liver disease, CKD, reproductive complications and **at least 13 cancer types**. Note the obesity-guideline lipid referral thresholds (TC >5.5, LDL-C >2.0, HDL-C <1.0, TG >2.0) differ from the Week 3 risk-based targets — use the source appropriate to the context.

Red flags — do not proceed with routine pharmacist-led weight management

- Suspected **eating disorder**
- Suicidal ideation or severe mental health concerns
- **Pregnancy or lactation**
- Severe or unexplained weight change
- Chest pain, significant breathlessness, persistent vomiting
- Possible undiagnosed endocrine disorder — unexplained fatigue, heat or cold intolerance, palpitations, menstrual irregularity, Cushing's features
- Suspected untreated OSA, or suspected MASLD / cardio-renal-metabolic disease

5. Building the plan — the 5As

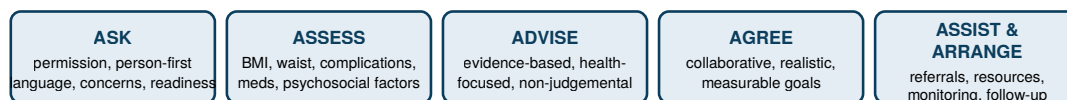


Figure 4. The 5As framework applied to obesity (RACGP SNAP Guide).

Lifestyle intervention is first-line for everyone: nutrition (energy deficit where required, better diet quality, sustainability), physical activity 150–300 min/week moderate or 75–150 min vigorous with muscle-strengthening at least twice weekly, reduced sedentary time, and behaviour change support (self-monitoring, food diaries, goal setting, problem solving, relapse prevention, motivational interviewing).

6. Dietary intensity — RED, LED and VLED

Diet	Energy	What it involves	When to use it
Reduced Energy Diet (RED)	~2000–4000 kJ/day <i>deficit</i> from usual intake	Modest deficit with balanced nutrition — more vegetables, fruit, legumes and wholegrains; less discretionary food, sugary drinks and energy-dense food	Appropriate first-line for many patients. Sustainable, no meal replacements needed, fits the Australian Dietary Guidelines
Low Energy Diet (LED)	~4200–5000 kJ/day <i>total</i> intake	Structured meal plans, partial meal replacement, or commercial programs	A step up when a reduced-energy diet alone hasn't achieved the target
Very Low Energy Diet (VLED)	<3300 kJ/day total intake	Nutritionally complete meal replacements (formulated products supplying full micronutrient requirements)	When lifestyle alone has failed, rapid weight loss is clinically indicated, in severe obesity, or preparing for bariatric surgery. Needs careful monitoring and multidisciplinary support

VLED and diabetes — the interaction to know. Patients on insulin or sulfonylureas are at particular risk of **hypoglycaemia** when calorie intake drops sharply, and those on **SGLT2 inhibitors** are at risk of **euglycaemic ketoacidosis** when carbohydrate intake falls markedly. Doses must be adjusted and glucose monitored closely before starting a VLED — this is a core pharmacist contribution to a weight-management plan.

7. Australian Obesity Management Algorithm

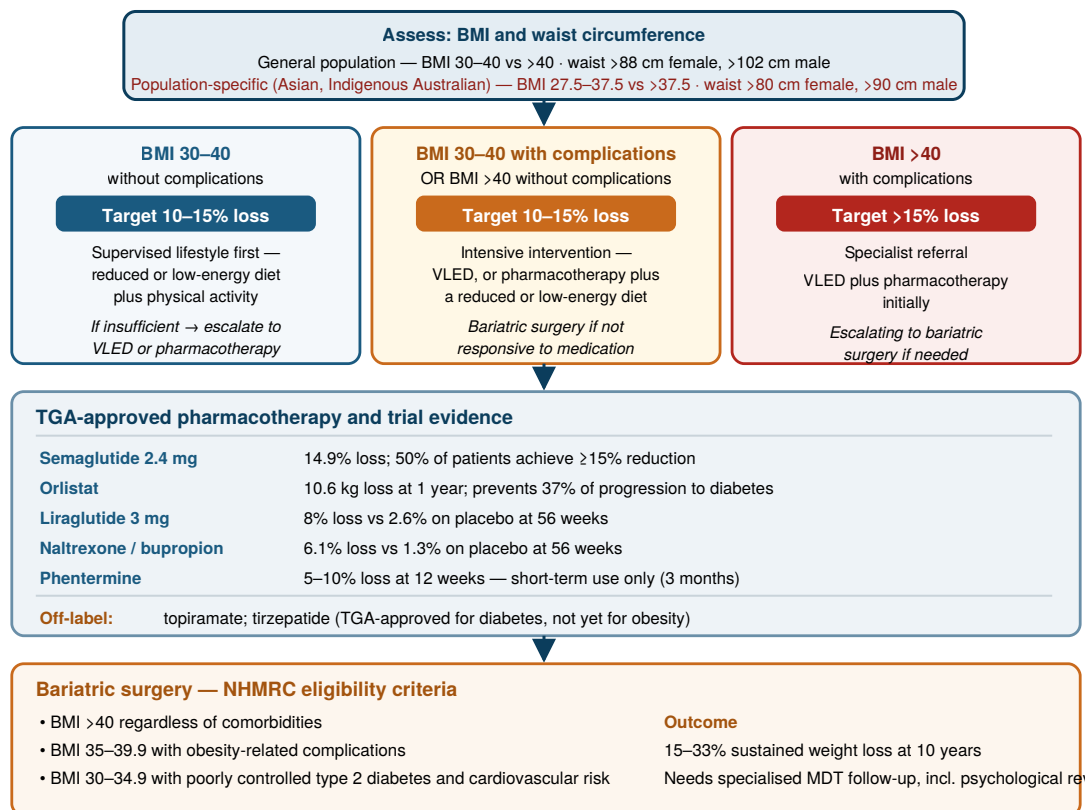


Figure 5. Australian Obesity Management Algorithm (Australian Diabetes Society, August 2025).

Note a discrepancy worth checking. The course notes describe the third pathway as "BMI >40, or BMI >35 with significant complications" targeting >15% loss. The published algorithm reserves the >15% target for **BMI >40 with complications**, placing BMI >40 without complications in the 10–15% group alongside BMI 30–40 with complications. Confirm which version your assessment uses.

8. Pharmacotherapy

Pharmacotherapy is an **adjunct to, not a replacement for, lifestyle change**. Consider escalation when lifestyle intervention has been properly implemented, the patient remains engaged, goals aren't being met, complications remain uncontrolled, or the patient wants to explore further options.

Drug	Dose	BMI threshold	Stopping rule	Key caution
Orlistat (in pharmacist scope)	120 mg up to 3× daily with main meals	>30, or >27 with CV risk factors	Skip the dose if a meal has no fat	Oily stools, flatus, faecal urgency — reduce dietary fat. Reduces vitamins A, D, E, K — take supplements 2 h apart. No evidence of CV outcome benefit
Phentermine	15 mg once daily at breakfast, max 40 mg	>30, or >25 with CV risk factors	Short term only, up to 12 weeks	Raises BP and heart rate. Contraindicated in uncontrolled hypertension, arrhythmias, cerebrovascular disease, MAOI use, drug-misuse history. Avoid in the elderly
Naltrexone / bupropion	Titrate over 4 weeks to 2 tablets twice daily	>30, or >27 with CV risk factors	Stop if <5% loss at 16 weeks	Contraindicated with chronic opioids, end-stage renal failure, hepatic impairment, pregnancy. Mean loss 4–5 kg at 1 year
Liraglutide	0.6 mg SC daily, +0.6 mg weekly, maintenance 3 mg daily	per obesity indication	Stop if <5% loss at 12 weeks on maintenance	GI effects. About 2 in 3 lose >5% of initial weight; plateau at 8–9 months
Semaglutide	0.25 → 0.5 → 1 → 1.7 → 2.4 mg weekly , 4 weeks per step (up to 7.2 mg if BMI >30)	>30, or >27 with CVD	Adolescents: reconsider if <5% BMI loss at 12 weeks on maintenance	About 12% more weight lost than placebo without diabetes, 6% with. The only agent here with proven CV outcome benefit

Tirzepatide (Mounjaro) — dual GIP/GLP-1 agonist, approved in Australia for type 2 diabetes, chronic weight management and OSA in specific groups. **Orforglipron** — an oral GLP-1 agonist in development, still under long-term study.

GLP-1 safety counselling

- **GI effects** peak during dose escalation — eat slowly, stop when comfortably full, smaller lower-fat meals, stay hydrated
- **Delayed gastric emptying** — tell any provider before surgery, sedation or a fasting procedure
- **Mental health** — monitor mood changes, worsening depression, suicidal ideation (causality unestablished)
- **Vision** — rare NAION reports; urgent review for sudden vision loss or field defect
- **Contraindicated in pregnancy** — discuss contraception and pre-pregnancy cessation with patients of reproductive age

Evidence beyond weight loss

- **SELECT** — semaglutide reduced major CV events in obesity with established CVD, with benefit appearing before substantial weight loss
- **FLOW** — semaglutide reduced kidney failure and death in CKD with type 2 diabetes
- **ESSENCE** — semaglutide significantly improved MASLD/MASH

These are why GLP-1 agents are increasingly framed as cardio-renal-metabolic therapies rather than weight-loss drugs.

9. Bariatric surgery

The most effective intervention for large sustained loss (around 25 kg, or 15–33% sustained at 10 years). **Selection:** generally BMI >40, or >35 with comorbidities such as diabetes or OSA; NHMRC criteria also allow BMI 30–34.9 with poorly controlled type 2 diabetes and cardiovascular risk. **Roux-en-Y gastric bypass** gives greater weight loss and glucose resolution than other

procedures, but higher immediate complication and 30-day mortality rates. **Gastric banding** has fewer deaths and shorter stays but more long-term complications (reflux, oesophageal dysmotility, band slippage or erosion) and more revisions. **Sleeve gastrectomy** is now used more often, including at lower BMIs. All need specialised multidisciplinary follow-up including psychological review — binge eating disorder is common in obesity and predicts less postoperative weight loss if unmanaged.

10. Monitoring and review

Obesity is chronic and relapsing, so follow-up is long-term rather than a single intervention. Review weight, waist, comorbidities, medicines, diet, activity, psychosocial wellbeing and adherence; reinforce behavioural strategies, address barriers, monitor pharmacotherapy adverse effects and adjust goals. **Frequency is individualised** — commonly every 2–4 weeks or monthly during active weight loss, then every 3–6 months in maintenance. The Queensland guideline uses monthly reviews with dispensing quantities aligned to the review schedule.

Plateaus are expected, not failure: energy requirements fall, hunger signals rise, metabolism adapts. Normalise it, review diet, activity, sleep, stress and adherence, and point to non-scale progress.

Queensland obesity protocol — the scope boundaries

Eligible: adults 18+, BMI ≥ 25 .

Do not treat, refer: pregnancy or lactation; **BMI >40** ; unmanaged comorbidities needing specialist care; previous bariatric surgery; currently on weight-gain-inducing medication; inadequate response ($<5\%$ loss after 12 weeks).

Treat and refer: diagnosed or suspected psychiatric or eating disorder; significant nutritional deficiency; CVD, asthma or COPD; **age over 65**; already receiving active medical weight-loss treatment elsewhere; signs of undiagnosed comorbidity.

Orlistat initiation: after Management Plan A (lifestyle) achieves $<5\%$ loss in 3 months, **and** either BMI >27 with CV risk factors or BMI >30 without. LFTs required during therapy.

Week 4 clinical nuggets

- **Ask about weight-gain-inducing medicines by name:** insulin, sulfonylureas, antipsychotics, antidepressants, mood stabilisers (lithium, sodium valproate), corticosteroids, some hormonal therapies — plus OTC products, protein powders and supplements.
- **Weight loss is not the only valid goal.** Weight maintenance, preventing further gain, better metabolic health, more activity or improved quality of life can all be legitimate outcomes.
- Phentermine is the odd one out in a CVD-focused course — it **raises blood pressure and heart rate**, so it's usually the wrong choice in exactly the patients this course is about. Note its lower BMI threshold (>25 with risk factors) too.
- **Weight is regained when GLP-1 treatment stops.** That's the pharmacological argument for treating obesity as an ongoing condition rather than a course of therapy — and a point patients need to hear before starting.
- **A 1% body weight reduction lowers systolic BP by about 1 mmHg** — a useful concrete number when motivating a patient who feels 5% sounds trivial.
- Waist-to-height ratio ≥ 0.5 needs no ethnic-specific adjustment, which makes it the most portable single measure across a diverse patient population.

Knowledge Check

Ten questions across Weeks 3 and 4. Answers follow on the next page.

1. A patient at high ASCVD risk has a baseline LDL-C of 3.6 mmol/L. After 6 weeks on atorvastatin 40 mg their LDL-C is 1.9 mmol/L. Have they met target? Show your reasoning.
2. Name the three circumstances in which a statin should be stopped, with the numerical thresholds where they apply.
3. A patient on maximally tolerated rosuvastatin has an LDL-C above target and triglycerides of 2.4 mmol/L. What is the next step for each abnormality?
4. An asymptomatic patient has a single HbA1c of 50 mmol/mol taken on a finger-prick meter in the pharmacy. Can you diagnose diabetes? What are the two problems with this result?
5. A 61-year-old with type 2 diabetes on metformin has an HbA1c of 62 mmol/mol, heart failure with reduced ejection fraction, and an eGFR of 38. Which second agent, and what happens to its glucose-lowering effect at that eGFR?
6. A patient on metformin, empagliflozin and gliclazide develops gastroenteritis with vomiting. What happens to each of the three medicines, and what single test must be done regardless of the blood glucose reading?
7. List four components of the Annual Cycle of Care and their frequencies.
8. A woman has a waist circumference of 91 cm and a BMI of 27. How do you classify each measurement, and which is the better predictor of cardiometabolic risk?
9. Under the Queensland obesity guideline, which of these patients must be referred rather than managed by the pharmacist?
(a) BMI 33 with hypertension (b) BMI 42 (c) 68-year-old with BMI 31 (d) BMI 28 with 4% weight loss after 12 weeks of lifestyle intervention
10. A patient with a BMI of 34 and no diabetes asks about semaglutide. Give the titration schedule, the maintenance dose, and the single most important thing to tell them about stopping treatment.

Answers

1. Yes. High risk requires a $\geq 50\%$ reduction *and* LDL-C < 1.8 — whichever is lower. Reduction = $(3.6 - 1.9) \div 3.6 = 47\%$, which falls short of 50%; but a 50% reduction would mean 1.8, and they've reached 1.9. Since the target is "whichever is lowest", and 50% of 3.6 is 1.8 while the absolute target is also 1.8, they are marginally above on both limbs — so strictly **not** at target, and adherence plus a dose increase or ezetimibe should be considered. This question rewards showing the working: the two limbs must both be satisfied.

2. Aminotransferase **persistently $> 3 \times$ ULN**; CK **$> 10 \times$ ULN**; or persistent unexplained muscle pain or weakness **even when CK is normal**. Rechallenge is possible after at least 4 weeks if the episode was mild and CK has normalised — consider a precipitant, a lower dose or a different statin. Never stop during ACS symptoms.

3. LDL-C above target → add ezetimibe (a further 15–25% reduction, and more effective than doubling the statin given the 80%-at-half-maximum-dose principle). **Triglycerides above target → add fenofibrate**, after optimising lifestyle and considering fish oil. Fenofibrate — not gemfibrozil — is the fibrate used with a statin, and needs FBC, LFTs and CK monitoring.

4. No. Two problems: (i) diabetes must be diagnosed on a **venous laboratory sample** — point-of-care capillary testing cannot be used; and (ii) in an **asymptomatic** patient a positive result must be **confirmed by a repeat measurement on a different day**, preferably using the same test. The value itself (50 mmol/mol, above the 48 threshold) is consistent with diabetes but doesn't establish it.

5. An **SGLT2 inhibitor** — it's the preferred add-on for heart failure and for CKD, and this patient has both. At an eGFR of 38 the **glucose-lowering effect is reduced** (it fades below eGFR 45 because the mechanism depends on renal glucose excretion), but the **heart failure and kidney benefits persist down to below eGFR 25**. Metformin can continue at this eGFR but the maximum dose should be reduced to 1 g daily.

6. Metformin — withhold (dehydration reduces clearance and raises lactic acidosis risk; it also worsens nausea). **Empagliflozin — withhold** (illness plus dehydration plus reduced carbohydrate intake raises DKA risk). **Gliclazide — continue as tolerated**, but withhold if glucose trends toward the lower range. Both withheld drugs restart only when the patient is eating and drinking normally and renal function is back to baseline. The essential test is **blood ketones** — a normal or only mildly raised glucose does not exclude euglycaemic DKA on an SGLT2 inhibitor, and urine ketones may be unreliable.

7. Any four of: HbA1c 3-monthly if unstable or 6-monthly if stable; blood pressure at every review; foot assessment every 1–12 months by risk; eye examination every 2 years; dental check every 12 months; uACR, eGFR and lipids every 12 months; weight and waist every 6 months; nutrition, activity, medication review, smoking status and psychosocial review every 12 months.

8. BMI 27 is **overweight** (25.0–29.9). A waist of 91 cm in a woman is **substantially increased** risk (≥ 88 cm; ≥ 80 is increased). **Waist circumference is the better predictor** of cardiometabolic risk because it reflects central and visceral adiposity, which BMI cannot distinguish from lean mass or peripheral fat. Waist-to-height ratio ≥ 0.5 is another useful measure and needs no ethnic adjustment.

9. (b) and (d) are do-not-treat/refer: BMI > 40 , and inadequate response defined as $< 5\%$ weight loss after 12 weeks. **(c)** is treat-and-refer — age over 65 is on the concurrent-referral list rather than an exclusion. **(a)** BMI 33 with hypertension can be managed, provided the hypertension is not unmanaged and requiring specialist care.

10. Wegovy, SC once weekly, escalating every 4 weeks: **0.25 → 0.5 → 1 → 1.7 → maintenance 2.4 mg** weekly (7.2 mg is an option after more than 4 weeks at 2.4 mg if baseline BMI > 30). If GI effects are significant, delay the increase rather than pushing through. The critical counselling point is that **weight is regained when treatment stops** — obesity is a chronic relapsing condition and this is long-term therapy, not a course. Also cover GI effects during escalation, the need to tell providers about delayed gastric emptying before surgery or sedation, and contraception if of reproductive age.

Sources: PHAR6302 course notes; Therapeutic Guidelines (Lipid modification; Classifying and diagnosing diabetes; Sick-day management for adults with diabetes); Australian Medicines Handbook, July 2026 (Dyslipidaemia, Statins, Type 2 diabetes, Metformin, Sulfonylureas, DPP-4 inhibitors, GLP-1 analogues, SGLT2 inhibitors, Orlistat, Phentermine, Naltrexone with bupropion); Heart Foundation Practical Guide to Pharmacological Lipid Management; Australian Type 2 Diabetes Glycaemic Management Algorithm (ADS, May 2026); Australian Obesity Management Algorithm (ADS, August 2025); Heart Foundation Obesity and CVD Consensus Statement; Queensland Community Pharmacy CVD Risk Reduction Program lipid, blood glucose and overweight/obesity protocols. Therapeutic Guidelines and AMH are subscription resources — confirm doses and combinations there.